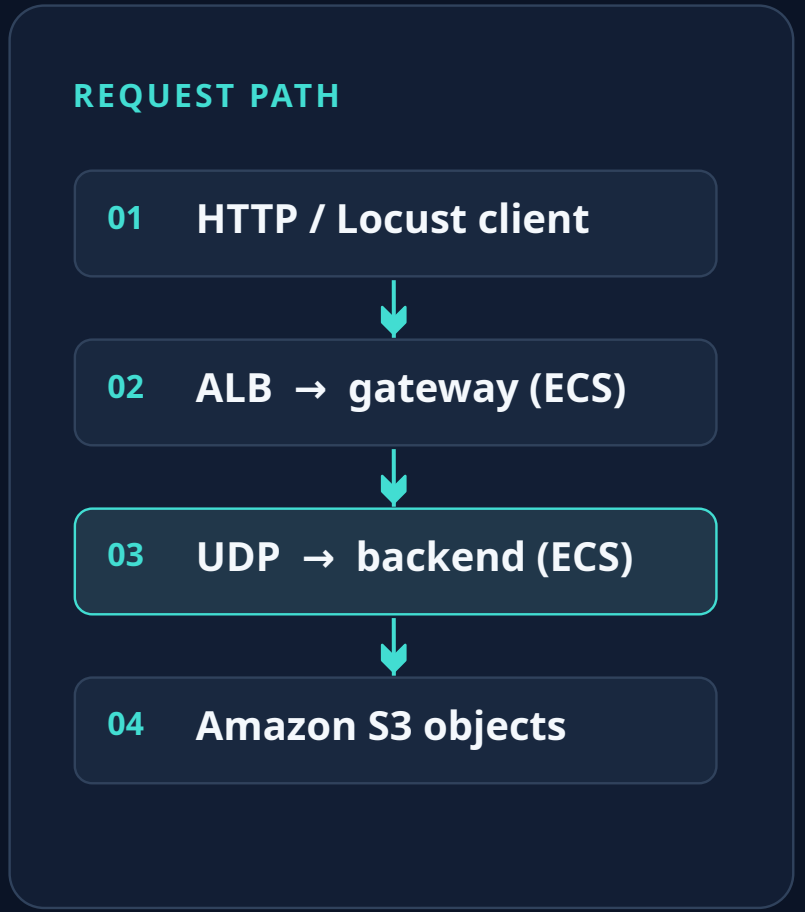
CS 6650 · BUILDING SCALABLE DISTRIBUTED SYSTEMS

Scaling a File-Transfer Service

Three empirical experiments on AWS: payload size, concurrent load, and UDP packet loss.

AWS ECS | LOCUST | UDP + S3



Empirical evaluation · 02 Nov 2026

NIKHIL K H | MIDTERM MASTERY PART 2

One system. Three controlled questions.

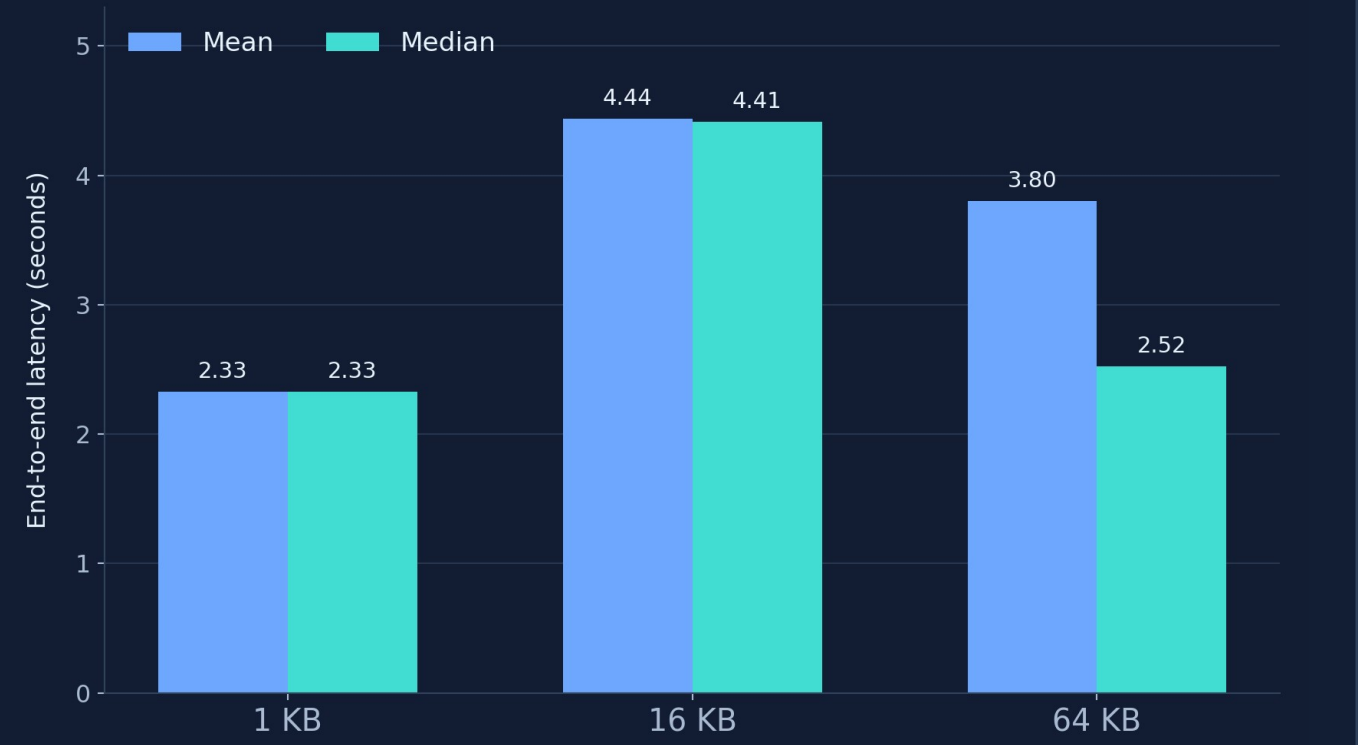
All measurements were collected through the HTTP file-download API.

		FIXED / CONTROLLED	OBSERVE
01	PAYLOAD SIZE **1 / 16 / 64 KB**	5 sequential reads per size	Mean + median latency; byte equality
02	CONCURRENCY **5 / 10 / 25 / 50 users**	16 KB; Locust; ~60 s	Failures, latency, completed requests
03	UDP SEND LOSS **0% / 5% / 10%**	5 users; 16 KB; receive loss 0%	Mean + p95; success / error counts

Correctness check HTTP 200 was not enough: scripts also checked the downloaded bytes.

Bigger files were not always slower.

Five sequential downloads per file size; no simulated loss; all responses verified.



15 / 15

valid file transfers

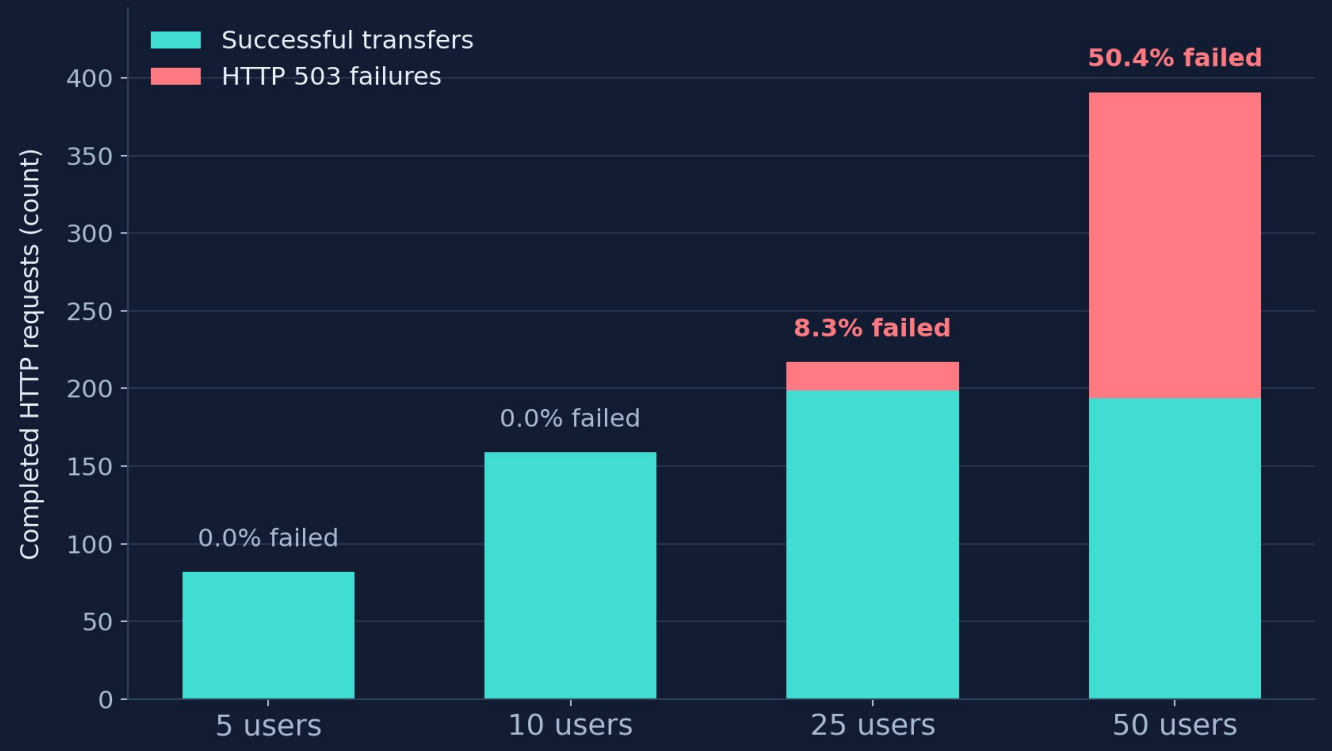
OBSERVATION

16 KB mean: 4.44 s
64 KB mean: 3.80 s

Latency varied within each size. Five samples cannot isolate UDP retries, queuing, or S3 delays.

Higher request rate ≠ useful throughput.

16 KB file; 5 → 50 Locust users; failures appear as HTTP 503.



50.4%

failed at 50 users (197 / 391)

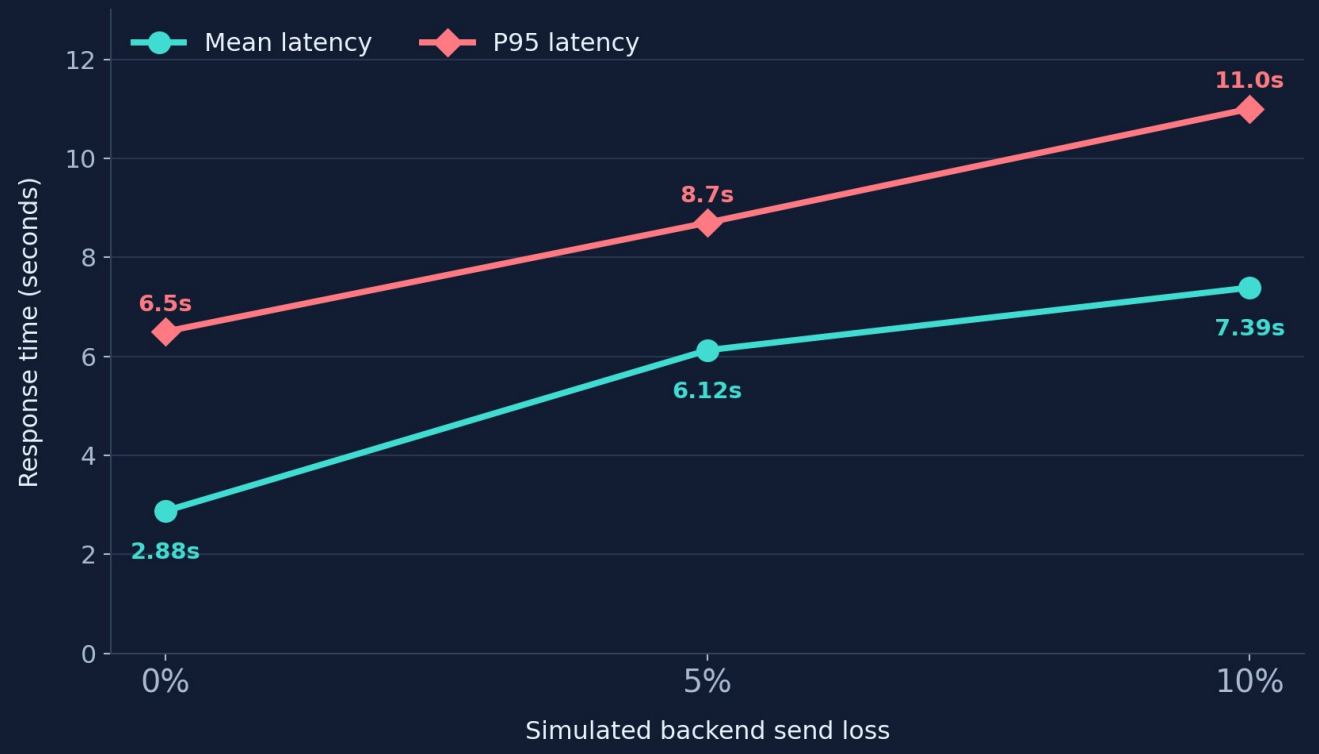
WHY THIS MATTERS

10 users: 0 failures
25 users: 18 HTTP 503s
50 users: 197 HTTP 503s

The successful-request count did not improve at 50 users (194) versus 25 (199).

Packet loss raised latency without observed corruption.

Simulated backend send loss of 0%, 5%, 10%; 5 Locust users; 16 KB file.



2.57×

mean latency at 10% vs 0%

RELIABILITY TRADEOFF

0% loss → 82 / 82 valid
5% loss → 40 / 40 valid
10% loss → 35 / 35 valid

Zero observed failures ≠ a guarantee.
Retry counts were not measured.

Scaling is a tradeoff, not one number.

What the experiments revealed - and what still needs measurement.

01

PAYLOAD VARIABILITY

Size alone did not explain latency.

Need repeated runs and server-side timings.

02

CAPACITY REJECTION

HTTP 503s reached 50.4% at 50 users.

Check gateway/backend inflight limits and LB logs.

03

LOSS PENALTY

10% send loss increased mean latency 2.57×.

Instrument UDP retries and deadlines.

NEXT EXPERIMENTS

Repeat runs at smaller load increments · inspect ALB/ECS errors and CPU · log retries and queue depths · compute sustained successful throughput